Srishti Gupta

Postdoctoral Researcher · CISPA Helmholtz Center for Information Security
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Secure deployment of machine learning: drift and out-of-distribution detection, robustness and deployment-grade evaluation across image, text, audio, code and malware.

Research Focus

I am interested in the intersection of machine learning robustness and security, with a focus on how AI systems behave once deployed in open, evolving and adversarial environments:

- **Adapting to distribution shift:** detecting out-of-distribution inputs and learning continually without forgetting or full retraining, for vision, malware and deepfake detection.
- **Exposing and mitigating failures of deployed AI:** adversarial and side-channel attacks, privacy leakage in memory-augmented LLM assistants and multi-agent systems, and trustworthiness gaps in audio language models.
- **Making AI security evaluation trustworthy:** uncovering methodological pitfalls, auditing LLM-as-a-judge evaluations and building generalization and robustness benchmarks.

Our NDSS 2026 work systematically identified pitfalls in LLM security research, and I co-organize MIRROR, the first workshop on meta-science in AI security research (IEEE SaTML 2027).

Education

Ph.D. in Artificial Intelligence (summa cum laude), National PhD Program in AI
Italy

Sapienza University of Rome, jointly with the University of Cagliari — *Feb. 2023 – May 2026*

- Thesis: *Robustness of Machine Learning Systems: under Distribution Shift & in Adversarial Environments*
- Advisors: Prof. Battista Biggio and Prof. Fabio Roli

M.Sc. in Electrical and Computer Engineering
Tucson, USA

University of Arizona — *Aug. 2019 – Aug. 2021*

- Thesis: *Impacts of Adversary Attacking Filter-Based Feature Selection Algorithms*
- Advisor: Prof. Gregory Ditzler
- GPA 3.75/4.0

B.Tech. in Electronics and Communication Engineering
New Delhi, India

Guru Gobind Singh Indraprastha University — *Aug. 2013 – Jun. 2017*

- GPA 3.8/4.0

Research Appointments

Postdoctoral Researcher
Saarbrücken, Germany

CISPA Helmholtz Center for Information Security · Group of Dr. Lea Schönherr — *May 2026 – present*

- Security, fairness and safety evaluation of audio language models
- Pitfalls in using LLMs for security research (NDSS 2026)
- Combining continual learning and OOD detection for autonomous model updates
- Efficient continual malware classification for real-world deployment

Visiting Researcher
Saarbrücken, Germany

CISPA Helmholtz Center for Information Security — *May – Aug. 2024 & Apr. – Dec. 2025*

- Continual learning against concept drift in malware classification
- Image-based adversarial attacks against anti-phishing engines

Graduate Research Assistant

Tucson, USA

University of Arizona

Jan. 2020 – Aug. 2021

- Machine Learning and Data Analytics Lab (2020 – 21): poisoning and evasion attacks on feature selection (IEEE Access 2021); real-time concept-drift detection on data streams
- Tactical Cyber Immune System, industry project (May – Dec. 2020): autoencoder-based insider-threat and zero-day detection; online signature-based intrusion detection
- Electroluminescence image analysis (Jan. – Aug. 2020): ML-based spatial fault detection in aged photovoltaic modules

Industry Experience

Associate Software Engineer

New York, USA

JPMorgan Chase · Securitized Products Group Data Analytics

Sep. 2021 – Jan. 2023

- Built automated ETL pipelines for daily and weekly agency mortgage data
- Migrated legacy servers to AWS (RDS, Aurora)
- Automated daily data pipelines feeding trader and research dashboards

Network Engineer

Gurgaon, India

National Institute of Information Technology (NIIT)

Jul. 2017 – Sep. 2018

- Trained engineers from client companies, including Nokia and Cisco, on L2/L3 protocols
- Configured remote routers and switches via the command line
- Received the Star of the Month award

Grants, Fellowships and Awards

2023 – 26	**PhD Scholarship,** Italian National PhD Program in AI, funded by PNRR (B53C22003860006)	*Italy*
2024	**Mobility Grant,** ELSA – European Lighthouse on Secure and Safe AI	*Europe*
2024	**Research Funding (contributor),** SafeAI project, University of Cagliari	*Italy*

Peer-Reviewed Publications

2026

NDSS **Chasing Shadows: Pitfalls in LLM Security Research**

Network and Distributed System Security Symposium (NDSS) · doi:10.60882/cispa.31861624

Jonathan Evertz, Niklas Risse, Nicolai Neuer, Andreas Müller, Philipp Normann, Gaetano Sapia, **Srishti Gupta**, David Pape, Soumya Shaw, Devansh Srivastav, Christian Wressnegger, Erwin Quiring, Thorsten Eisenhofer, Daniel Arp, Lea Schönherr

IJCV **Out-of-Distribution Detection for Continual Learning: Design Principles and Benchmarking**

International Journal of Computer Vision (IJCV)

Srishti Gupta, Riccardo Balia, Daniele Angioni, Fabio Brau, Maura Pintor, Ambra Demontis, Alessandro Sebastian, Salvatore Mario Carta, Fabio Roli, Battista Biggio

TIFS **Regression-aware Continual Learning for Android Malware Detection**

IEEE Transactions on Information Forensics and Security (TIFS) · doi:10.1109/TIFS.2026.3714132

Daniele Ghiani, Daniele Angioni, Giorgio Piras, Angelo Sotgiu, Luca Minnei, **Srishti Gupta**, Maura Pintor, Fabio Roli, Battista Biggio

T-ITS **Security of Deep Reinforcement Learning for Autonomous Driving: A Survey**

IEEE Transactions on Intelligent Transportation Systems (T-ITS) · doi:10.1109/TITS.2026.3740283

Ambra Demontis, **Srishti Gupta**, Maura Pintor, Luca Demetrio, Kathrin Grosse, Hsiao-Ying Lin, Chengfang Fang, Battista Biggio, Fabio Roli

DAI **Over-parameterization and Adversarial Robustness in Neural Networks: An Overview and Empirical Analysis**

Discover Artificial Intelligence, Springer Nature · doi:10.1007/s44163-026-01621-5

Srishti Gupta, Zhang Chen, Luca Demetrio, Fabio Brau, Xiaoyi Feng, Zhaoqiang Xia, Antonio Emanuele Cinà, Maura Pintor, Luca Oneto, Ambra Demontis, Battista Biggio, Fabio Roli

2025

PR **Buffer-free Class-Incremental Learning with Out-of-Distribution Detection**
Pattern Recognition · doi:10.1016/j.patcog.2025.112441
Srishti Gupta, Daniele Angioni, Maura Pintor, Ambra Demontis, Lea Schönherr, Battista Biggio, Fabio Roli

ICML-W **BUILD: Buffer-free Incremental Learning with OOD Detection for the Wild**
ICML Workshop on Foundation Models in the Wild
Srishti Gupta, Daniele Angioni, Ambra Demontis, Lea Schönherr, Battista Biggio, Fabio Roli

2021

Access **Gradient-Based Attacks on Information-Theoretic Feature Selection**
IEEE Access, vol. 9 · doi:10.1109/ACCESS.2021.3105555
Srishti Gupta, et al.

Theses and Software

Ph.D. **Robustness of Machine Learning Systems: under Distribution Shift & in Adversarial Environments**
Ph.D. thesis, Sapienza University of Rome, defended May 18, 2026 · hdl.handle.net/11573/1768180
Srishti Gupta

M.Sc. **Impacts of Adversary Attacking Filter-Based Feature Selection Algorithms**
M.Sc. thesis, University of Arizona, 2021 · ProQuest
Srishti Gupta

Code **BUILD: buffer-free continual learning with OOD detection**
github.com/srish01

Code **Continual and OOD learning for robot vision**
PyTorch, Avalanche · github.com/srish01

Talks and Presentations

2025	**Cue-Rich Evaluation of Speech and Trustworthiness,** EurIPS 2025 Workshop on Foundations of Language Model Security	*Copenhagen, Denmark*
2024	**BUILD: Buffer-free Incremental Learning with OOD Detection for the Wild,** ICML Workshop on Foundation Models in the Wild	*Vienna, Austria*

Teaching

2026	**Trustworthy Machine Learning,** Guest lectures and tutorials, University of Hamburg	*Hamburg, Germany*
2017 – 18	**Corporate Instructor,** L2/L3 networking protocols for engineers from Nokia and Cisco, NIIT	*Gurgaon, India*

Supervision and Mentoring

Current

2026 – 27	**Esha Afzal,** M.Sc. research, co-supervisor · Projection attacks on audio language models	*In progress*
2026 – 27	**Mubeen Afzal,** M.Sc. thesis, supervisor · Robustness of AI-generated harnesses for bug-detecting fuzzers	*In progress*
2026 – 27	**Teodor Vasilev,** B.Sc. project, supervisor · Chain-of-thought monitoring via random paraphrasers in RLHF	*In progress*

Past

2026	**Alex Dugast**, M.Sc. thesis, supervisor · Multi-agent system security	*Paper in review*
2026	**Mete Cem Turan**, B.Sc. internship, supervisor · Continual malware detection	*Paper in review*
2026	**Onur Aray**, B.Sc. project, supervisor · Continual deepfake detection	*Demo*

Academic Service

Organization and Chairing

2027	**Co-Organizer**, MIRROR: 1st Workshop on Meta-Science in AI Security Research @ IEEE SaTML · mirror-workshop.io	*Reykjavík, Iceland*
2025	**Discussion Chair**, Privacy and Alignment, Foundations of Language Model Security @ EurIPS · llmsec-eurips.github.io	*Copenhagen, Denmark*

Program Committees and Reviewing

2026	**Artifact Evaluation Committee**, Network and Distributed System Security Symposium (NDSS)
2026	**Reviewer**, IEEE Secure and Trustworthy Machine Learning (SaTML)
2026	**Reviewer**, ACM Web Conference (WWW)
2026	**Reviewer**, Conference on Language Modeling (COLM)
2026	**Reviewer**, ACM Workshop on Artificial Intelligence and Security (AISec)
2026	**Reviewer**, IEEE Transactions on Information Forensics and Security (TIFS)
2025 – 26	**Reviewer**, Pattern Recognition (journal)
2025	**Reviewer**, ECML-PKDD Workshop on Secure and Trustworthy AI (STAI)
2025	**Reviewer**, ICLR Workshop on Test-Time Updates (TTU)
2025	**Reviewer**, Structural, Syntactic, and Statistical Pattern Recognition (S+SSPR)

Challenges

2025	**Participant**, SAND: Speech Analysis for Neurodegenerative Diseases, IEEE ICASSP 2026 Grand Challenge · sand.icar.cnr.it

Skills and Languages

Research Areas	Continual learning · out-of-distribution detection · adversarial machine learning · LLM and agent security · side-channel attacks · privacy and contextual integrity · deepfake and AI-generated content detection · malware detection under concept drift · audio language models · evaluation methodology and benchmarking
Models	LLMs and LLM agents (single- and multi-agent), audio language models, vision transformers, CLIP, diffusion models, CNNs
ML Frameworks	PyTorch, Hugging Face Transformers, Avalanche, scikit-learn, scikit-multiflow, TensorFlow
Engineering	Python, Bash, Git, Linux, SQL, R, MATLAB, AWS (RDS, Aurora)
Languages	Hindi (native) · English (fluent) · Italian (B1) · French (B1) · German (A2)

References

Prof. Dr. Battista Biggio
University of Cagliari, Italy
battistabiggio.github.io

Prof. Dr. Fabio Roli
University of Genoa, Italy
saiferlab.ai/people/fabioroli

Dr. Lea Schönherr
CISPA Helmholtz Center for Information Security, Germany
leaschoenherr.me

Dr. Gregory Ditzler
Lockheed Martin AI Center; formerly University of Arizona
gditzler.github.io